Test 2 is Friday. Today: every problem type from your Unit 2 quizzes.

- **Turn in Quiz 9** now.
- Test 2 has **15 questions, 200 points, on 4 pages**, built from the skills on Quizzes 5 to 9: about 10% Section 4.2, 30% Section 4.3, 30% polygons, 10% DMS and 20% circles.
- For each type: I work a model, then you try one. Answers appear when we check.
- Twelve questions are single skills: the four logarithm questions (3 to 6) are worth 15 points, the rest 10. Three have two parts worth 20. Show your work for every question.

Printed on Test 2

- $A = P(1 + r/n)^{nt}$ and $A = Pe^{rt}$
- change of base: $\log_b a = \ln a \div \ln b$
- Heron: $s = (a + b + c)/2$, $A = \sqrt{(s(s-a)(s-b)(s-c))}$

From memory

- $\log_b x = y \iff b^y = x$; $\ln x = c \iff x = e^c$
- trapezoid: $A = \frac{1}{2}(b_1 + b_2)h$
- DMS: $D + M/60 + S/3600$; inscribed angle = ½ arc
- radians = degrees × $\pi/180$; $s = r\theta$; $A = \frac{1}{2}r^2\theta$
- $v = r\omega$, with ω = rpm × 2π

SECTION 4.2 · COMPOUND INTEREST (QUIZ 5)

$A = P(1 + r/n)^{nt}$

r as a decimal · n = 2 semiannually, 4 quarterly, 12 monthly (Test 2 states n, e.g. "monthly (n = 12)") · round to the nearest cent at the end

Model: \$2,500 at 3.5% compounded monthly for 6 years

Setup: $A = 2500(1 + 0.035/12)^{12 \cdot 6} = 2500(1 + 0.035/12)^{72}$

$A \approx \$3{,}083.25$

You try: \$7,000 at 5.2% compounded quarterly for 9 years

Setup: $A = 7000(1 + 0.052/4)^{36}$

$A \approx \$11{,}143.92$

$A = Pe^{rt}$

"Compounded continuously" means use e · multiply r × t first · round to the nearest cent

Model: \$1,500 at 6.2% compounded continuously for 7 years

Setup: $A = 1500e^{0.062 \cdot 7} = 1500e^{0.434}$

$A \approx \$2{,}315.13$

You try: \$9,000 at 3.8% compounded continuously for 11 years

Setup: $A = 9000e^{0.418}$

$A \approx \$13{,}670.29$

The log is the exponent, even a fraction

$$b^y = x \iff \log_b x = y$$

Model

(a) $4^{3/2} = 8$ in log form → $\log_4 8 = 3/2$

(b) $\log_9 27 = 1.5$ in exponential form → $9^{1.5} = 27$

You try

(a) $25^{3/2} = 125$ in log form → $\log_{25} 125 = 3/2$

(b) $\log_4 32 = 2.5$ in exponential form → $4^{2.5} = 32$

$\ln x = c$ and $b^x = c$

$\ln x = c \to x = e^c$ · $b^x = c \to x = \log_b c = \ln c \div \ln b$ change of base is printed · four decimal places

Model

$\ln x = 2.5 \to x = e^{2.5} \approx 12.1825$

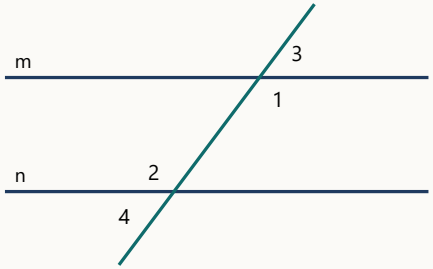
$3^x = 50 \to x = \log_3 50 = \ln 50 \div \ln 3 \approx 3.5609$

You try

$\ln x = 4.2 \to x = e^{4.2} \approx 66.6863$

$6^x = 150 \to x = \ln 150 \div \ln 6 \approx 2.7965$

Equal or supplementary?



m ∥ n. Alternate interior (1 and 2), alternate exterior (3 and 4) and corresponding angles are **equal**. Same-side interior angles **add to 180°**.

Model: alternate interior angles (2x + 27)° and (4x − 13)°

Equation: $2x + 27 = 4x - 13 \rightarrow x = 20 \rightarrow \text{angle} = 2(20) + 27 = 67°$

You try: same-side interior angles (2x + 20)° and (4x + 10)°

$(2x + 20) + (4x + 10) = 180 \rightarrow x = 25 \rightarrow$ angles $70°$ and $110°$

Matching angles are equal; matching sides are proportional

△ABC ~ △DEF means AB ↔ DE, BC ↔ EF, AC ↔ DF. Keep one triangle on top: AB/BC = DE/EF.

Model: angle A = 38°, angle B = 75°; AB = 6, BC = 10, DE = 9

$$\text{Angle F} = \text{angle C} = 180 - 38 - 75 = 67°$$

$$\text{EF: } 6/10 = 9/x$$

$$6x = 90 \rightarrow x = 15$$

You try: angle A = 49°, angle C = 76°; AB = 14, BC = 21, DE = 12

$$\text{Angle E} = \text{angle B} = 55°$$

$$\text{EF: } 14/21 = 12/x$$

$$14x = 252 \rightarrow x = 18$$

Two areas: trapezoid from memory, Heron printed

Trapezoid: $A = \frac{1}{2}(b_1 + b_2)h$, h the perpendicular height, not the slanted side · Heron: $s = (a + b + c)/2$, $A = \sqrt{(s(s - a)(s - b)(s - c))}$

Model: trapezoid with parallel sides 10 m and 18 m, slanted side 8 m, height 7 m; triangle with sides 5, 12, 13

$$\text{Trapezoid: } \tfrac{1}{2}(10 + 18)(7) = 98 \text{ m}^2$$

$$s = (5 + 12 + 13)/2 = 15$$

$$A = \sqrt{(15 \cdot 10 \cdot 3 \cdot 2)} = \sqrt{900} = 30$$

You try: trapezoid with parallel sides 6.5 cm and 11.5 cm, slanted side 5 cm, height 4 cm; triangle with sides 7, 15, 20

$$\text{Trapezoid: } 36 \text{ cm}^2$$

$$s = 21$$

$$A = \sqrt{(21 \cdot 14 \cdot 6 \cdot 1)} = \sqrt{1764} = 42$$

DMS both ways

decimal → DMS: × 60, then × 60 again (nearest second) · DMS → decimal: D + M/60 + S/3600 (four places)

Model

$54.2875° \to 0.2875 \times 60 = 17.25′$; $0.25 \times 60 = 15″ \to 54°17′15″$

$31°12′45″ \to 31 + 12/60 + 45/3600 = 31.2125°$

You try

$98.4561° \to 98°27′22″$

$17°53′24″ \to 17.8900°$

Half the arc; radians before s = rθ

central angle = its arc · inscribed angle = ½ arc · radians = degrees × π/180 · $s = r\theta$

Model: radius 8, central angle AOC = 120°

Inscribed angle ABC $= 120 \div 2 = 60°$

$120 \times \pi/180 = 2\pi/3$ rad

$s = 8(2\pi/3) = 16\pi/3 \approx 16.76$

You try: radius 10, central angle AOC = 84°

Inscribed angle $= 42°$

$84° = 7\pi/15$ rad; $s = 14\pi/3 \approx 14.66$

$A = \frac{1}{2}r^2\theta$, and $v = r\omega$

θ in radians · ω = rpm × 2π rad per minute · $v = r\omega$ in ft/min

Model: sector r = 10, $\theta = 2\pi/5$; wheel r = 2.5 ft at 160 rpm

$A = \frac{1}{2}(100)(2\pi/5) = 20\pi \approx 62.83$ square units

$\omega = 160(2\pi) = 320\pi$; $v = 2.5(320\pi) = 800\pi \approx 2513.3$ ft/min

You try: sector r = 6, $\theta = 5\pi/4$; wheel r = 1.5 ft at 210 rpm

$A = 22.5\pi \approx 70.69$ square units

$v = 630\pi \approx 1979.2$ ft/min

Where points are usually lost

- Rate as a percent (5.2) instead of a decimal (0.052).
- Wrong n: quarterly is 4, semiannually is 2, monthly is 12.
- Moving the base to the wrong side in log form.
- Change of base as ln(c ÷ b) instead of ln c ÷ ln b.
- Using the trapezoid's slanted side as its height.
- Rounding in the middle of a calculation.
- Solving for x but not finding the angle.
- Mismatched sides in a similar-triangle proportion.
- Skipping the semiperimeter, or a negative factor in Heron.
- Doubling an inscribed angle instead of halving the arc.
- Dividing seconds by 60 instead of 3600.
- Using degrees in $s = r\theta$ or $A = \frac{1}{2}r^2\theta$.
- Leaving ω in rpm, or using a diameter as the radius.

Before Friday

Tonight and Thursday

- Write every formula that isn't printed from memory until you can do it without looking.
- Redo every "you try" from today without notes.
- Work the Test 2 practice PDF on the course page, then check it.
- Take the Test 2 readiness check on the review page until you reach 80%.

Friday, October 30

- Test 2, in class on paper: 15 questions, 200 points, 50 minutes
- Printed: interest formulas, change of base and Heron's formula; everything else from memory
- Scientific calculator
- Show your work inside each question's box. A correct answer with no work gets no credit.